

Lecture 20

ChE 467:596-012 Polymer Rheology

Saad A Khan

of

Recap & Moving Forward

A. Dilute Systems

1. spheres (Einstein eqn.,)

2. fibers (brief)

concept of rotation

concept of $Pe^{\#}$

B. Non-dilute Systems

1. Interparticle forces & potential energy

2. Interpret rheology

of

Non-dilute Systems

1. Interparticle forces & potential energy

- **Attractive forces (dispersion)**
 - **van der Waals**
induced dipoles, possible in all systems
- **Repulsive forces**
 - **electrostatic**
 - **steric**

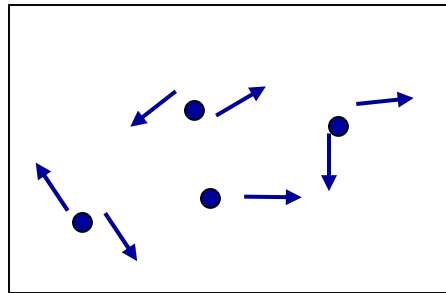
2. Interpret rheology (in terms of)

- **Hard spheres (Brownian)**
- **Stable system**
- **Flocculated system**

of

Non-dilute Suspensions-Brownian Motion

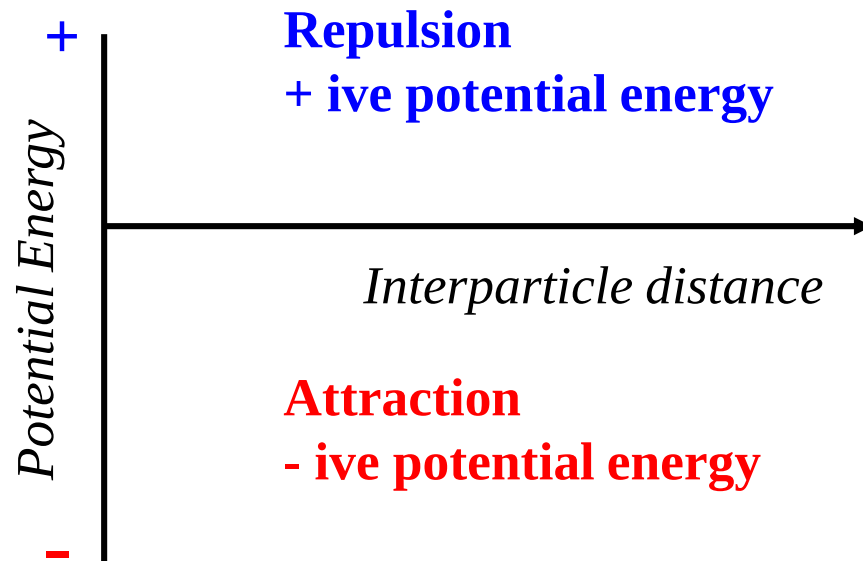
- *Tends to randomize things*
- *Concerned with translational diffusion coefficient*



of

Non-dilute Suspensions-Interparticle forces

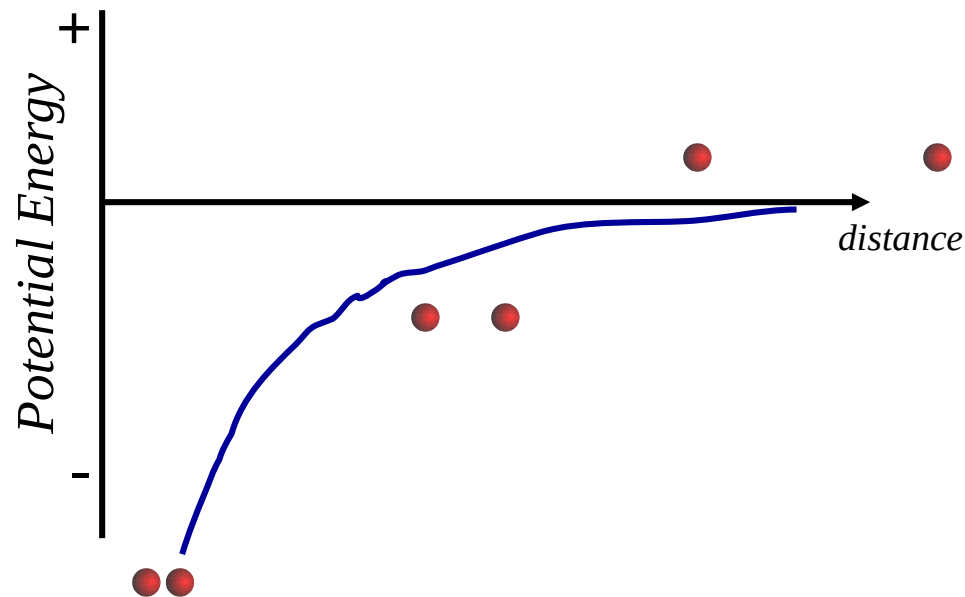
Interparticle forces (attractive or repulsive) are often looked from the stand point of potential energy



of

Non-dilute Suspensions-Interparticle forces

Schematic for generic attractive forces between two particles



of

van der Waals Interactions

- *Attractive dispersion forces*
- *Exists because of induced dipole of neighboring particles*



Potential Energy $\Rightarrow V_D$
Dispersion forces prevail

$$V_D = -A_m H_g$$

$A_m \Rightarrow$ Hamaker constant

$H_g \Rightarrow$ dependent on distance and geometry of particles

of

Hamaker Constant

- A_m is a measure of attraction due to induced dipole

$$A_m = (A_p - A_i)^{1/2}$$

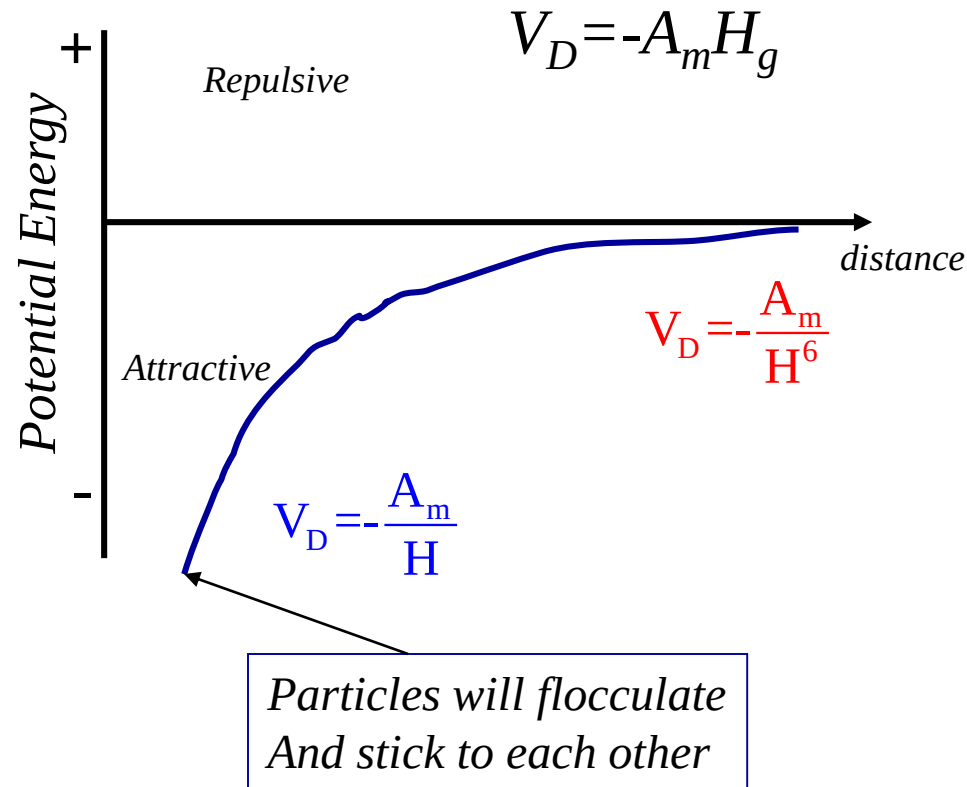
particle solvent

- H_g is dependent on the distance (H)

$$H_g \propto \frac{1}{H} \quad \text{as } H \rightarrow 0$$
$$H_g \propto \frac{1}{H^6} \quad \text{as } H \rightarrow \infty$$

of

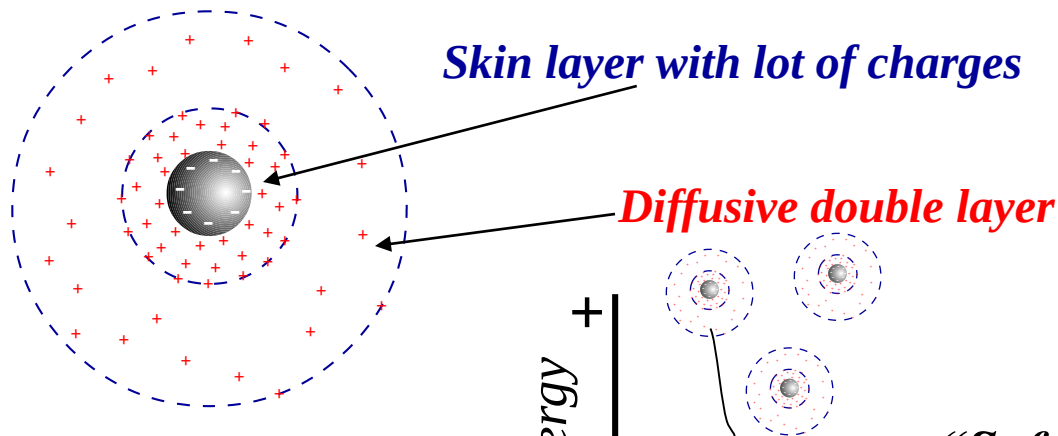
Hamaker Constant, cont'd



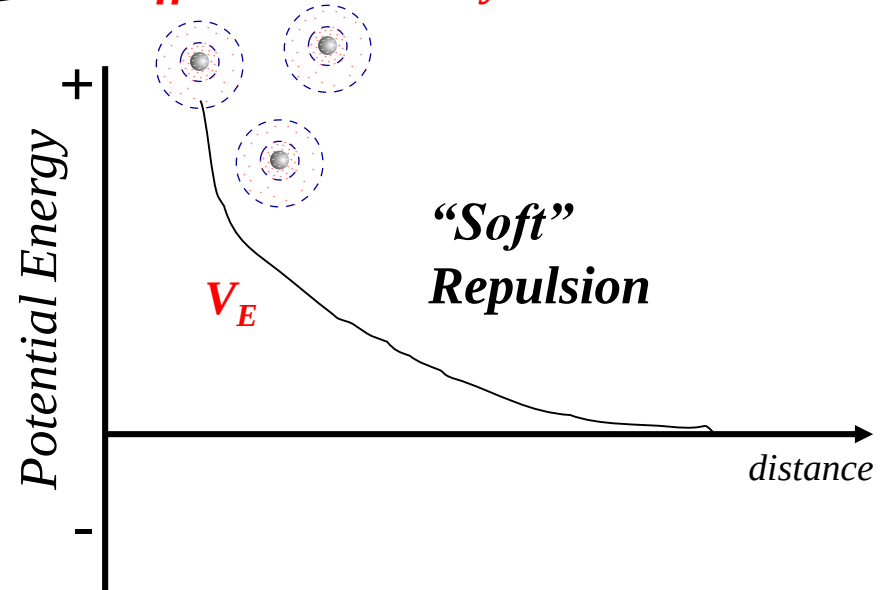
of

Repulsive Interactions

Electrostatic: Between charged particles

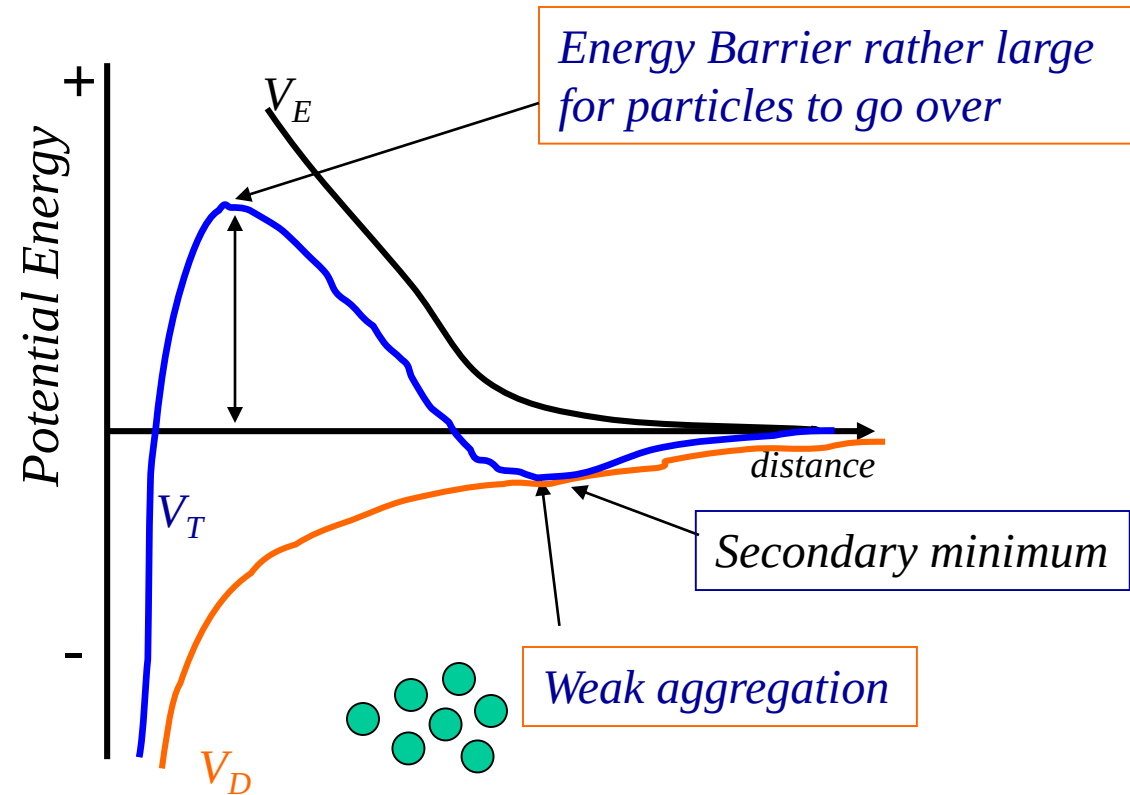


Take several of these particles and put in liquid medium



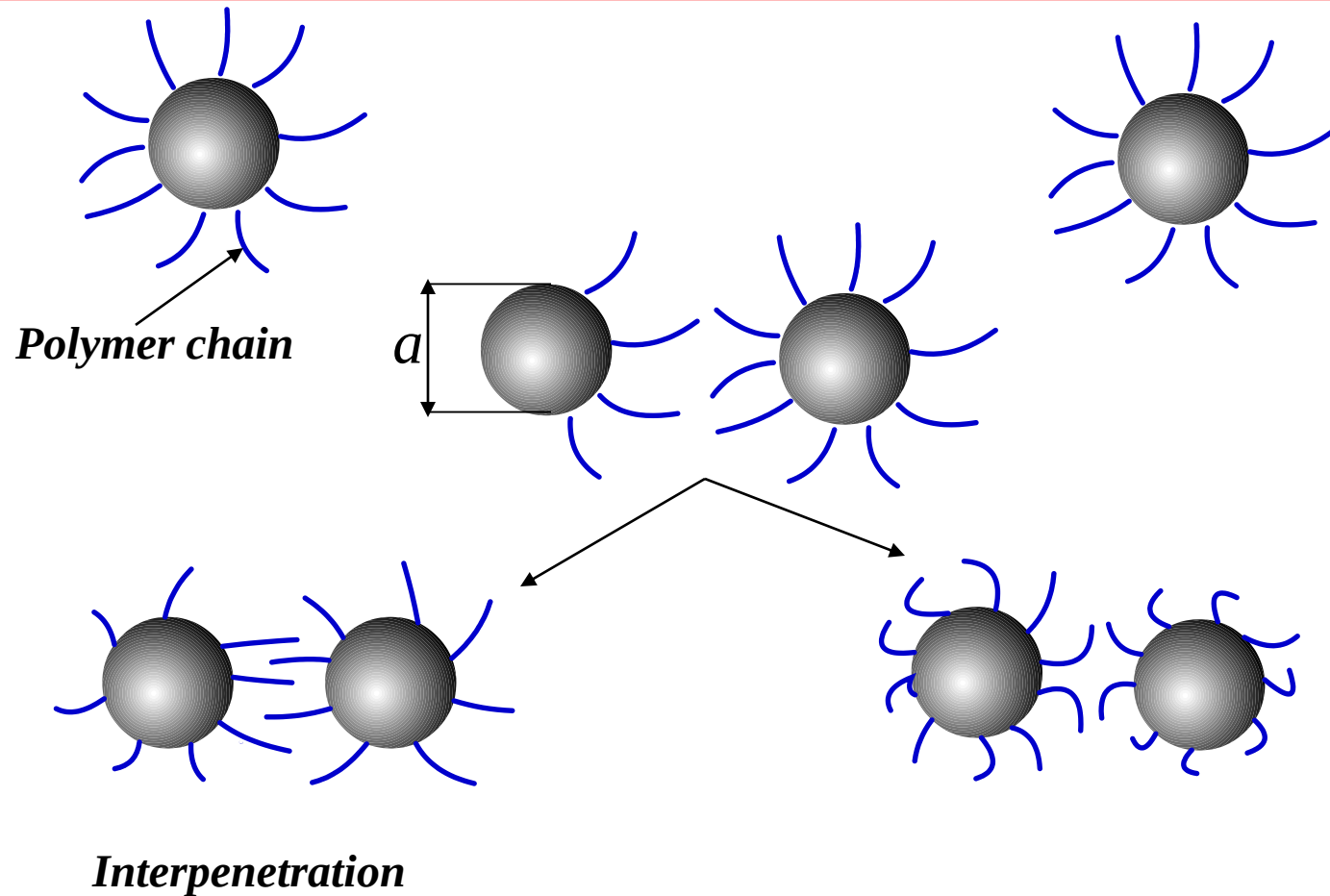
of

Electrostatic with Dispersion Interactions



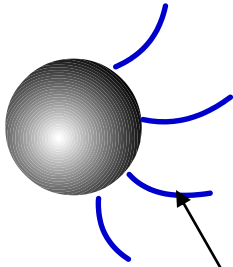
of

Steric Repulsion

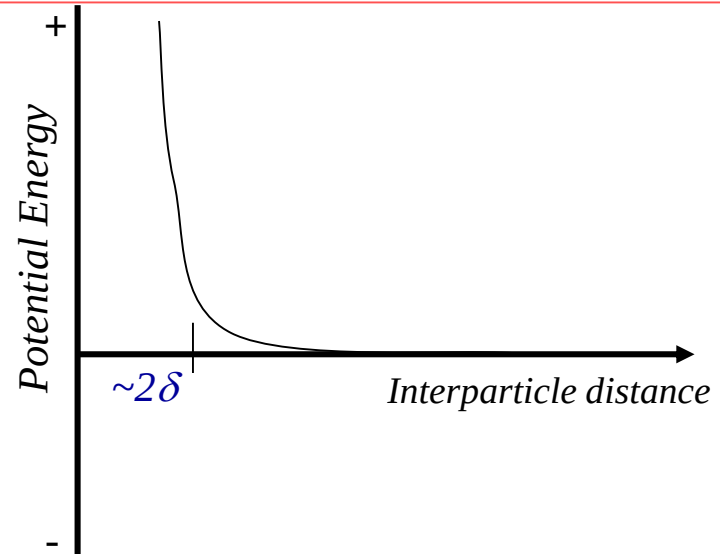


of

Steric Repulsion, cont'd



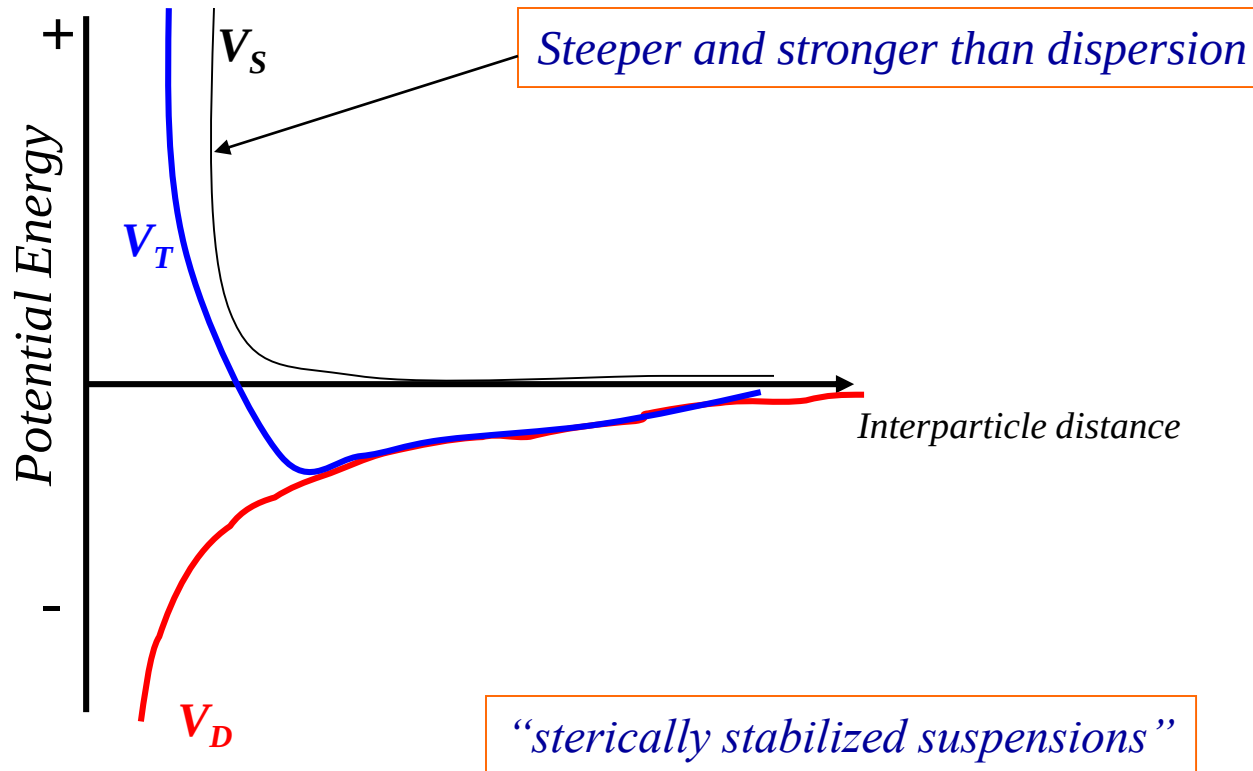
Chemically attached or physically adsorbed



- Very steep repulsive potential because particle cannot overlap $\Rightarrow$ chains on surface
- “Short range” interaction $\Rightarrow$ v. short distance over which force acting

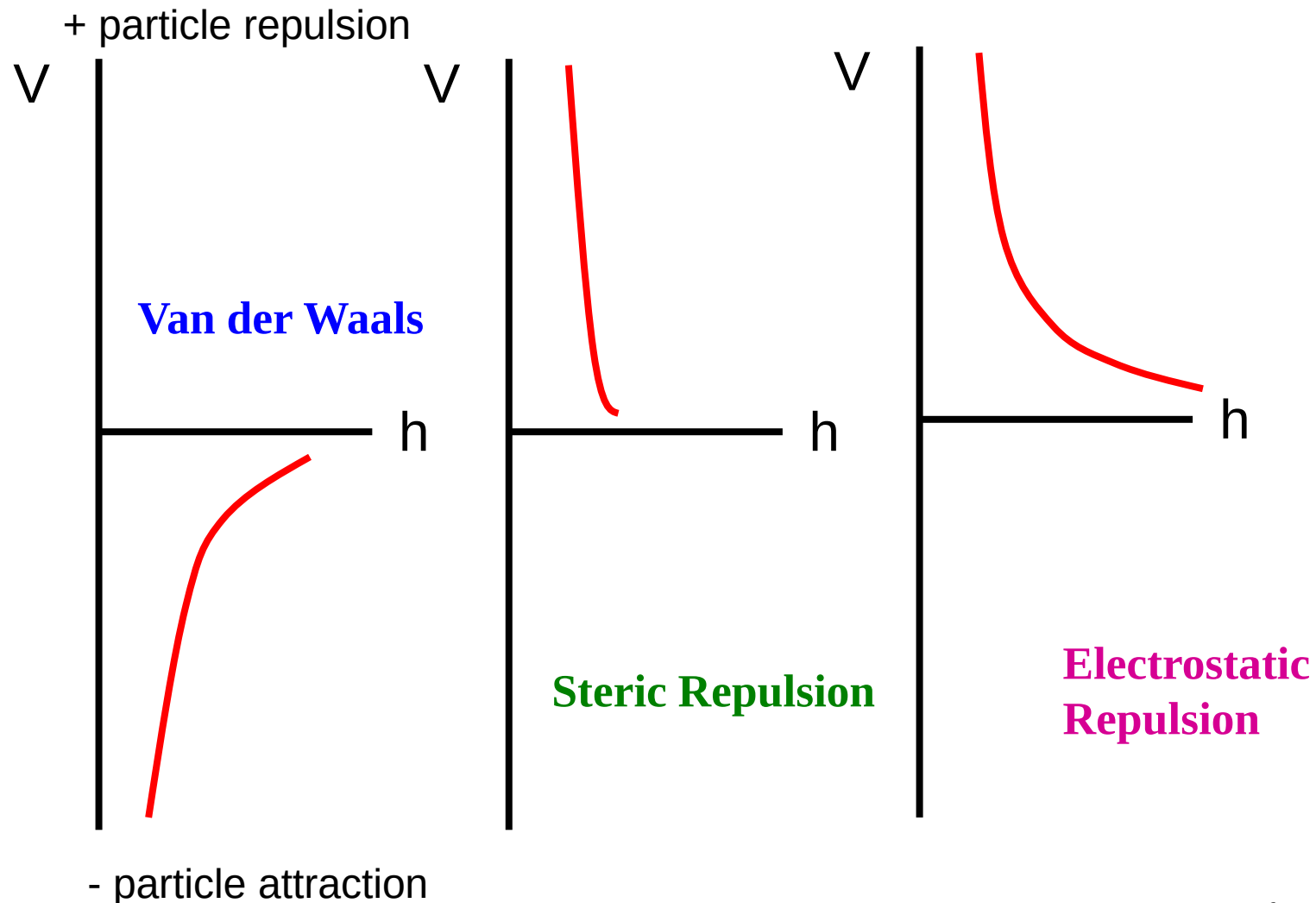
of

Steric with Dispersion Interactions



of

Interparticle Potential vs. Separation



of

Other attractive forces besides vdW?

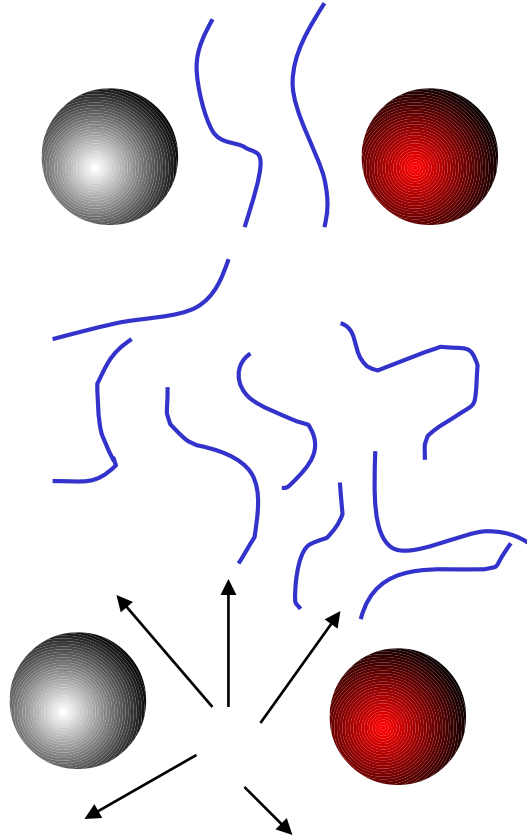
1. Hydrogen bonding

2. Depletion flocculation

3. Sticky chains (reverse of steric stabilization)

of

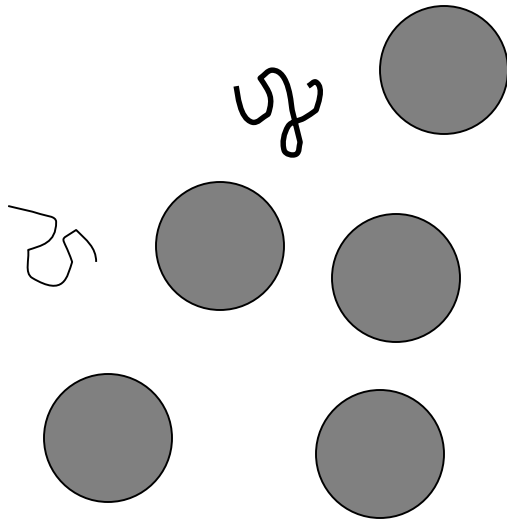
Depletion Flocculation



Polymer chains move out from between particles

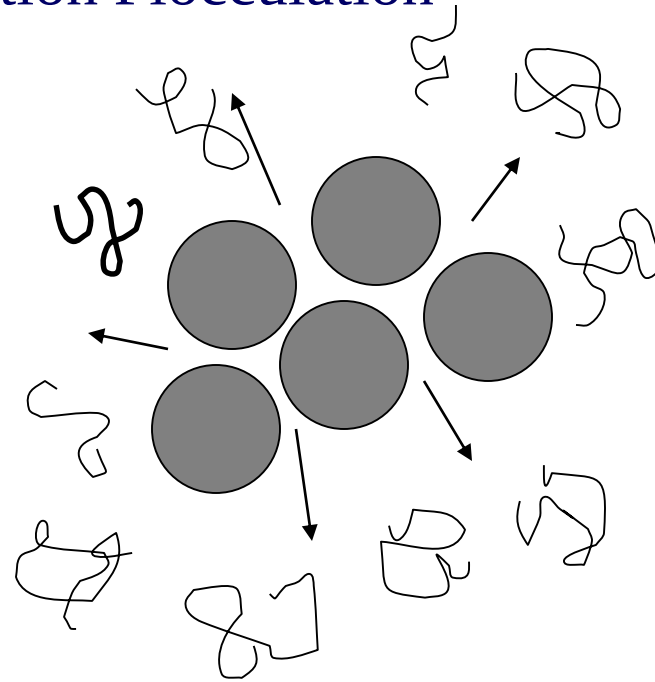
of

Non-adsorbing polymer - Depletion Flocculation



Low polymer concentration

Stable Dispersion

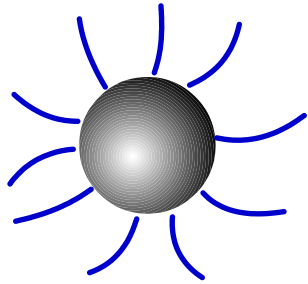


High polymer concentration
Flocculation by volume
restriction mechanism

$$\phi > \phi_p^+$$

of

Steric Forces



$$V_s = \frac{akT \left(\frac{1}{2} - \chi \right) \delta^2}{H_s v_s}$$

δ = thickness of layer

χ = Flory-Huggins interaction parameter

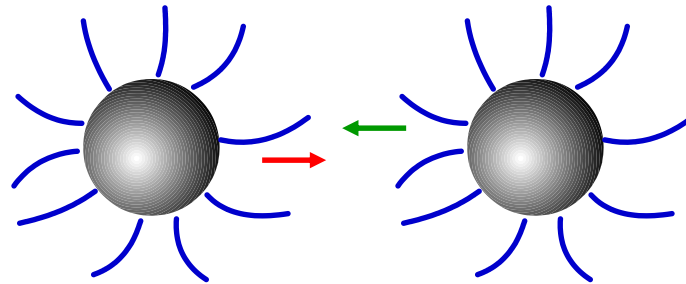
$H_s \sim$ function of interparticle distance

$V_s \sim$ volume terms

$\chi > 1/2$, V_s is negative



Attractive potential



of

Scaling Factors

$$\text{Re}^\# = \frac{\rho a v}{\mu} \quad \text{Pe}^\# = \frac{\eta_s \dot{\gamma} a^3}{kT} \equiv \frac{\text{Hydrodynamic forces}}{\text{Brownian forces}}$$

When colloidal forces present, we also have

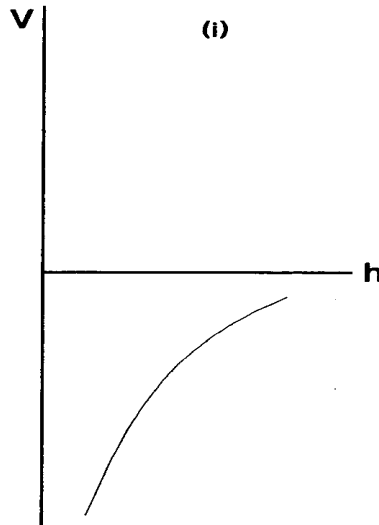
$$\frac{\text{Dispersion forces}}{\text{Brownian forces}} = \frac{A}{kT}$$

$$\frac{\text{Electrostatic forces}}{\text{Brownian forces}} \sim \frac{\epsilon \psi_o^2 a}{kT}$$

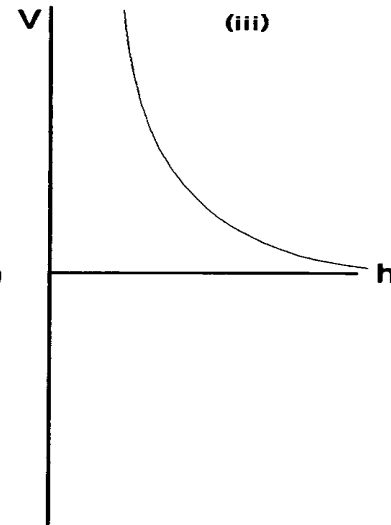
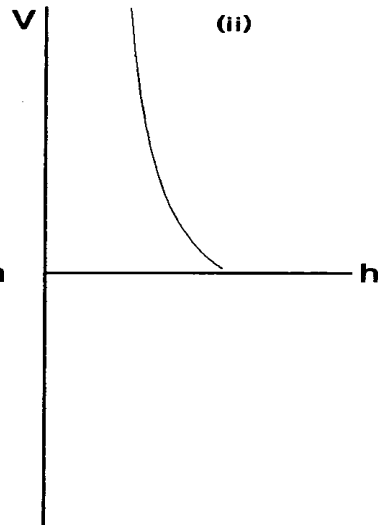
$$\frac{\text{Steric forces}}{\text{Brownian forces}} \sim \frac{(\frac{1}{2} - \chi) a \delta^2 kT}{v_s kT} \sim \frac{(\frac{1}{2} - \chi) a \delta^2}{v_s}$$

of

(+) Particle Repulsion

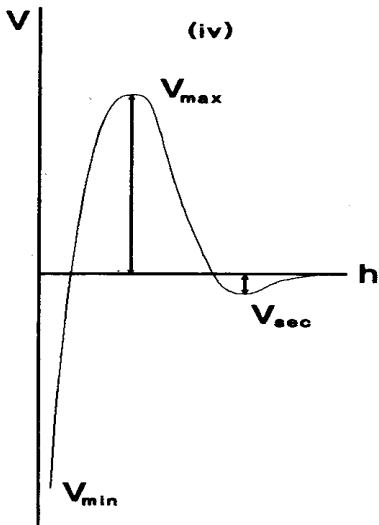


(-) Particle Attraction

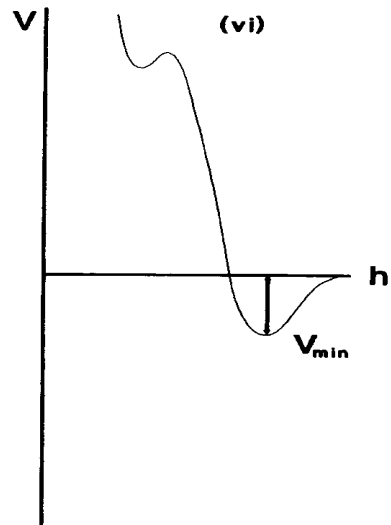
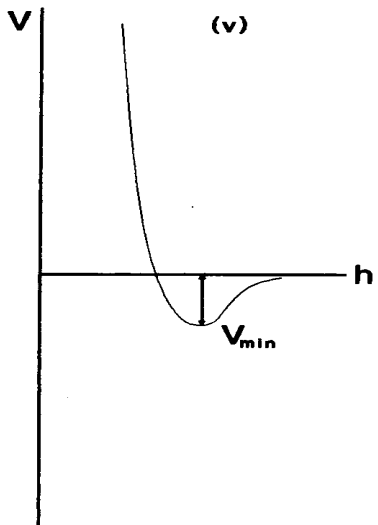


What interactions are occurring in each of these graphs?

(+) Particle Repulsion



(-) Particle Attraction



of

Rheological Behavior

*What does viscosity versus shear rate look like?
What is the frequency spectrum of G' and G'' ?
How does volume fraction affect these rheology?*

- *3 possible scenarios*
 - *Brownian hard spheres*
 - *Stable suspensions*
 - *Flocculated suspensions*

of

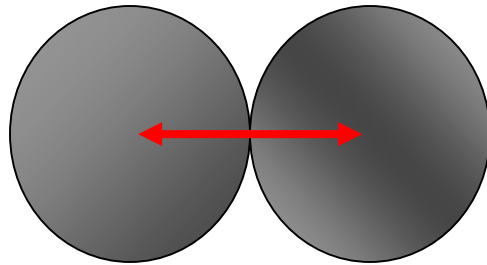
Classification of Dispersion Systems

- “Hard Sphere” systems
 - interparticle interactions screened
- Electrostatically stabilised systems
 - charged particles, electrokinetic phenomena
- Sterically stabilised systems
 - adsorbed polymeric species
 - stability thermodynamic in nature
- Flocculated systems
 - attractive interparticle forces predominate

of

“Hard” and “Soft” Particles

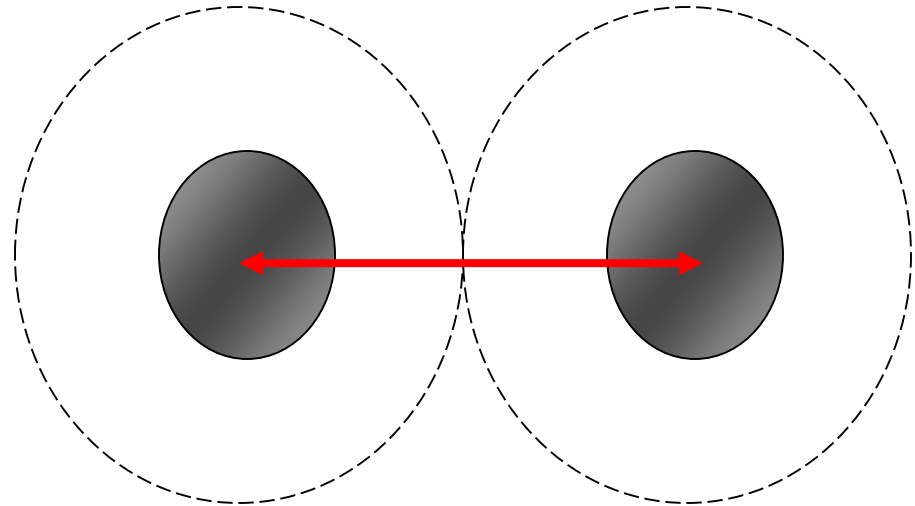
Hard



Colloidal interactions
screened out

Presence of neighbouring
particles experienced only
at close approach
i.e. interparticle separation
 $= 2 \times \text{particle radius}$

Soft



Adsorbed polymer or
electrical double layer

Presence of neighbouring
particles experienced at
greater interparticle separations_{of}

Rheological Behavior

- *3 possible scenarios*
 - *Brownian hard spheres*
 - *Stable suspensions*
 - *Flocculated suspensions*

Brownian hard spheres

- *Neglect dispersion, electrostatic, and steric interactions*
- *Interparticle forces: Brownian, hydrodynamic*
- *Resort to Peclet[#]*
$$Pe^{\#} = \frac{\eta_s \dot{\gamma} a^3}{kT}$$
- *At higher particle concentration, more appropriate to replace η_s with η (suspension)*

of

Rheological Behavior

3 possible scenarios

- *Brownian hard spheres*
- *Stable suspensions*
- *Flocculated suspensions*

Brownian hard spheres

- *Neglect dispersion, electrostatic, and steric interactions*
- *Interparticle forces: Brownian, hydrodynamic*
- *Resort to Peclet[#]*
- *At higher particle concentration, more appropriate to replace η_s with η (suspension)*

Brownian hard spheres—2 regimes

- *Dilute regime – already covered most of it*
- *Non-dilute regime*

of

Dilute Hard Sphere Systems

Particle volume fraction, $\phi < 0.1$

$$\frac{\eta}{\eta_s} = 1 + 2.5\phi$$

Classical Einstein relationship in which the relative viscosity is a simple linear function of the particle volume fraction

Note - the particle **weight** or **size** does not appear in the above equation.

The factor 2.5 is associated with the intrinsic viscosity of the system, $[\eta]$ -

$$[\eta] = \lim_{\phi \rightarrow 0} \left| \frac{(\eta - \eta_s)}{\eta_s \phi} \right| = \lim_{\phi \rightarrow 0} \left| \frac{\eta_r - 1}{\phi} \right|$$

of

Dilute Hard Sphere Systems

Effect of particle geometry:

$[\eta] = 2.5$ only holds for spherical particles

Generally, for all particle shapes $\eta_r = 1 + [\eta] \phi$

$[\eta]$ of polymers vs dispersions

Do you see any difference between the $[\eta]$ of polymers and dispersions?

Polymers- $\eta_{\text{rel}} = 1 + [\eta]c + K'[\eta]^2 c^2 + \dots \rightarrow [\eta] = \lim_{c \rightarrow 0} \frac{\eta - \eta_s}{c\eta_s}$

$[\eta]$ of dispersions– no units; for polymers – unit of inverse concentration

Class
Ended
Here

of

Non-dilute Hard Sphere Systems

Deviation from Einstein equation, in that higher order terms in ϕ are required to describe the viscosity - volume fraction relationship:

$0.1 < \phi < 0.2$ - Bachelor equation, spherical particles:

$$\eta_r = 1 + 2.5\phi + 6.2\phi^2$$

More generally: $\eta_r = 1 + \alpha\phi + \beta\phi^2$

What happens at higher concentration/vol. fraction?

Brownian Hard Spheres

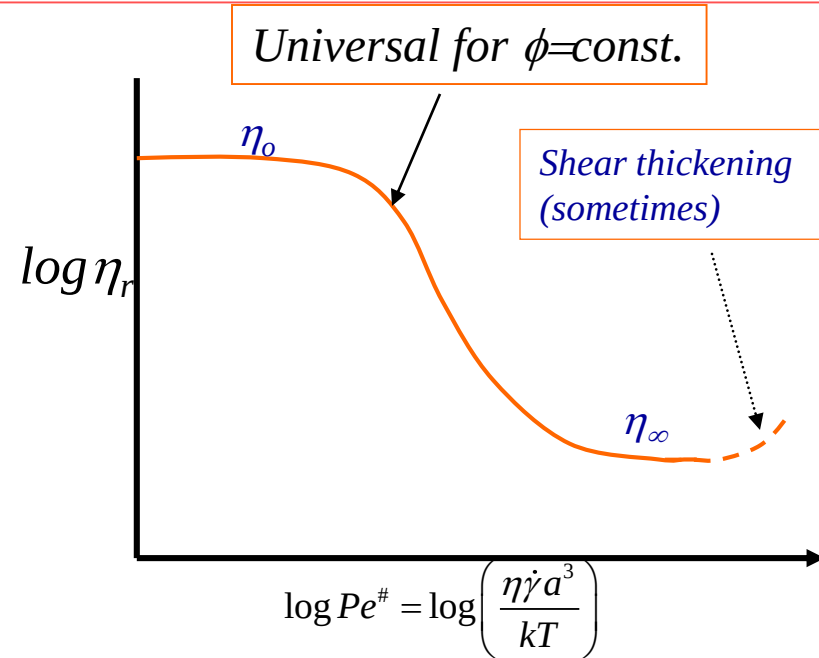
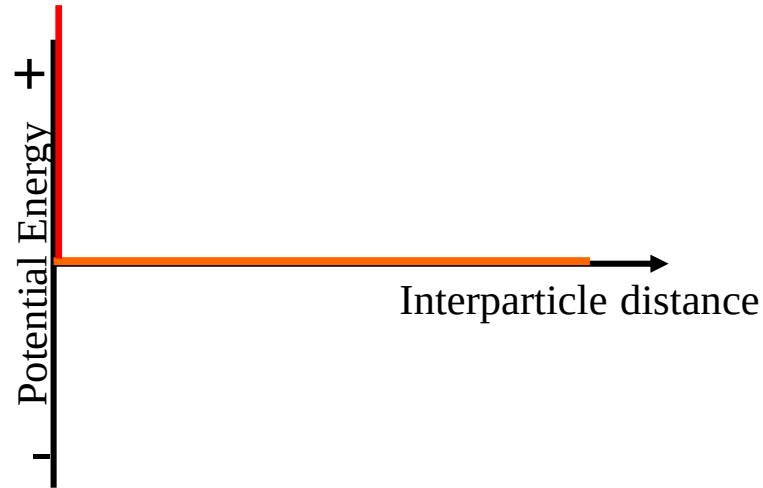
Let's look at the details of hard spheres

Brownian hard spheres

- ***Neglect dispersion, electrostatic, and steric interactions***
- ***Interparticle forces: Brownian, hydrodynamic***
- ***Resort to Peclet[#]***
$$Pe^{\#} = \frac{\eta_s \dot{\gamma} a^3}{kT}$$
- ***At higher particle concentration, more appropriate to replace η_s with η (suspension)***

of

Brownian Hard Spheres



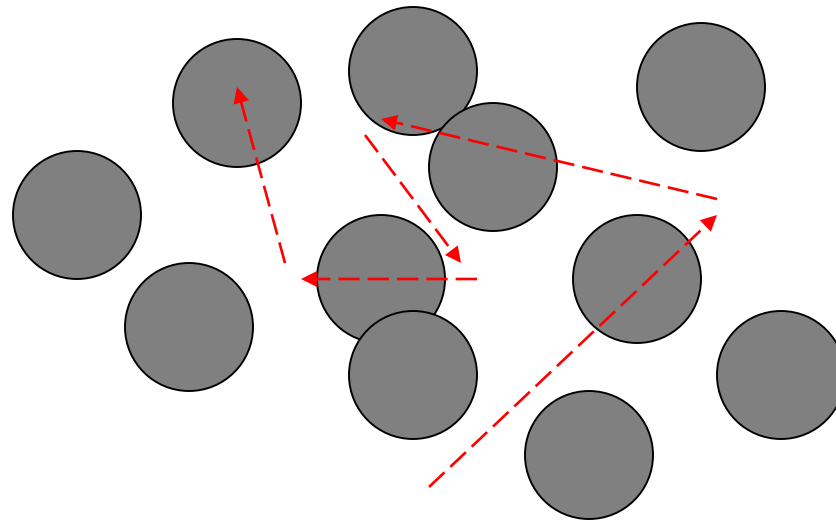
- *Universal behavior for monodisperse system*
Independent of τ , η_s , and particle size
- *Polydispersity can affect behavior*

of

Non-dilute Hard Sphere Systems

Effect of Shear Deformation

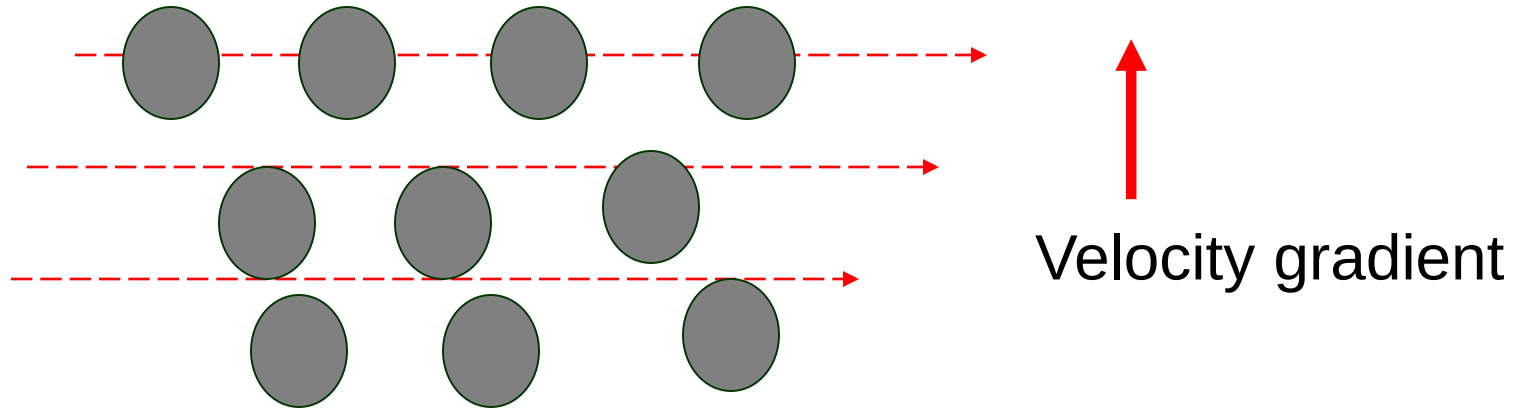
Low Rates of Deformation



**Randomising effect of thermal diffusion (“Brownian motion”)
leads to state of high relative viscosity**

of

High Rates of Deformation

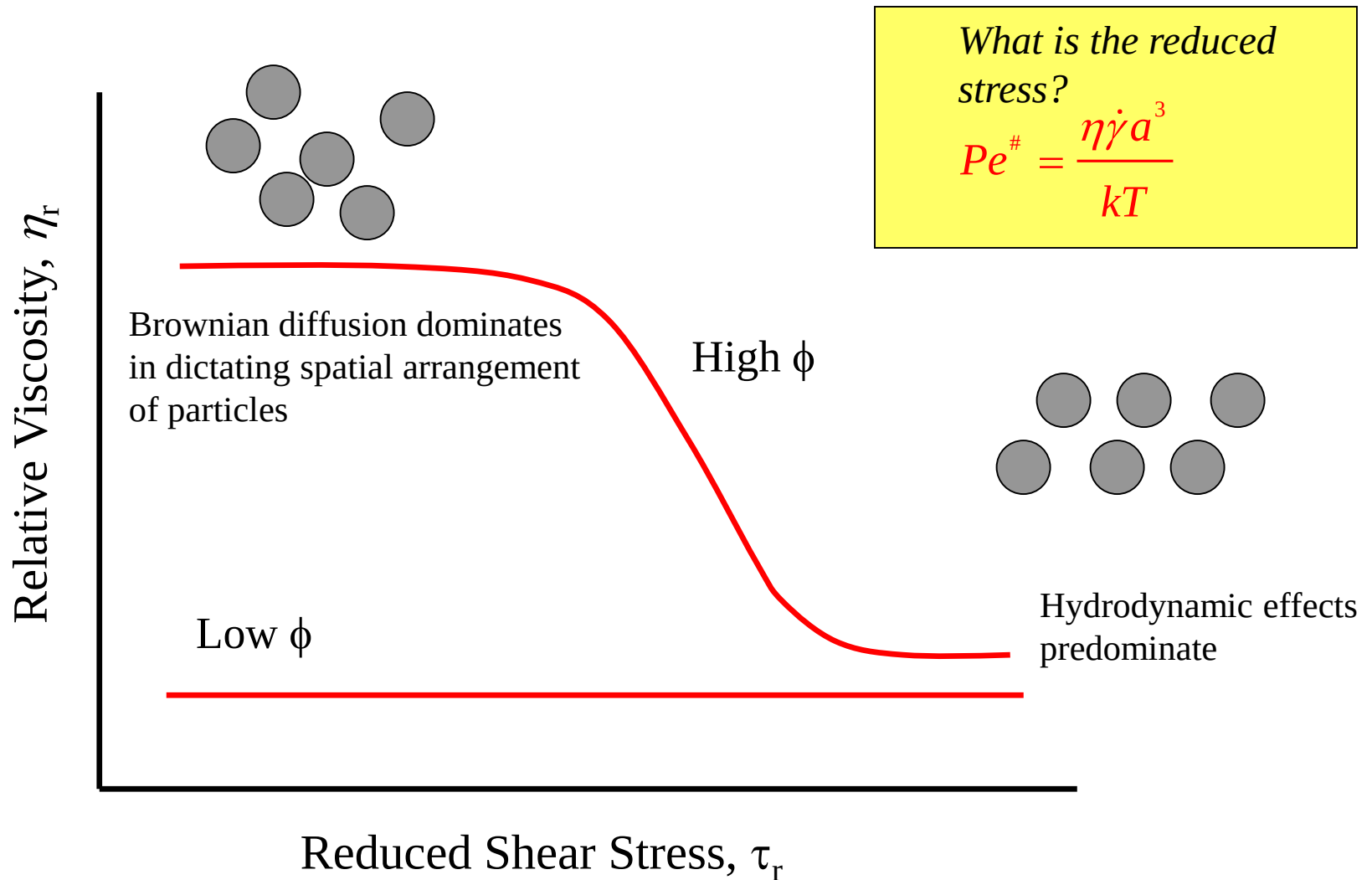


Predominance of hydrodynamic effects which arise as a consequence of the imposed shear deformation

Spatially, the particles adopt a lamellar arrangement which corresponds to a state of low relative viscosity

of

Generalised Viscosity - Stress Plot



of

Brownian Hard Spheres, cont'd

Of interest are:

- $\eta_o = f(\phi)$ ϕ = volume fraction of solid
- $\eta_\infty = f(\phi)$ ϕ = volume fraction of solid

Use Krieger-Dougherty relation:

$$\eta_{rx} = \left[1 - \frac{\phi}{\phi_{\max}} \right]^{-2.5\phi_{\max}}$$

relative

$$\begin{aligned} x=0 & \Rightarrow \eta_o \\ x=\infty & \Rightarrow \eta_\infty \end{aligned}$$

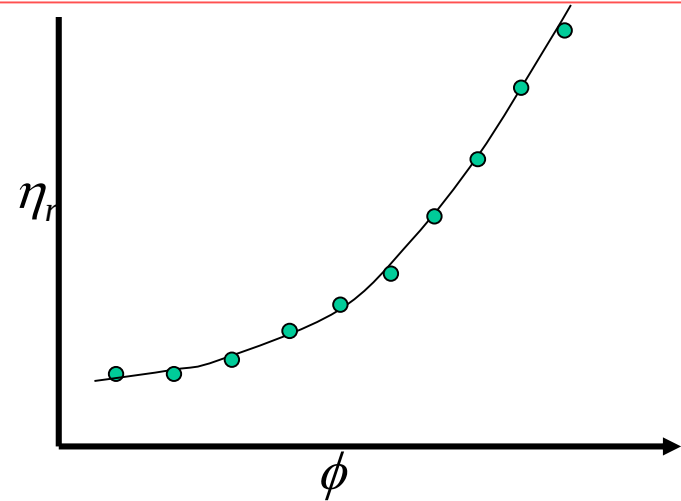
$\phi_{\max}$ = max packing fraction

$$\begin{aligned} \dot{\gamma} \rightarrow 0, & \quad \phi_{\max} = 0.64 - 0.63 \\ \dot{\gamma} \rightarrow \infty, & \quad \phi_{\max} = 0.71 \end{aligned}$$

of

Krieger-Dougherty Relation

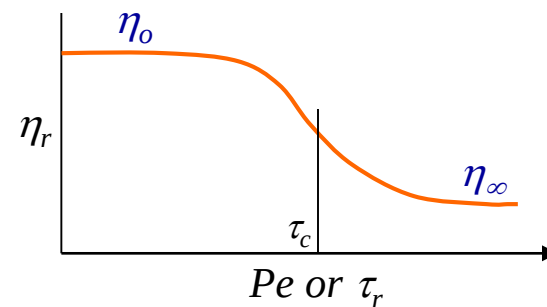
$$\eta_{\text{rx}}^{\text{relative}} = \left[1 - \frac{\phi}{\phi_{\text{max}}} \right]^{-2.5\phi_{\text{max}}}$$



Krieger also proposed:

$$\frac{\eta - \eta_{\infty}}{\eta_o - \eta_{\infty}} = \frac{1}{1 + \tau_r / \tau_c}$$

τ_c = reduced shear stress for which viscosity reaches average value between two Newtonian plateau

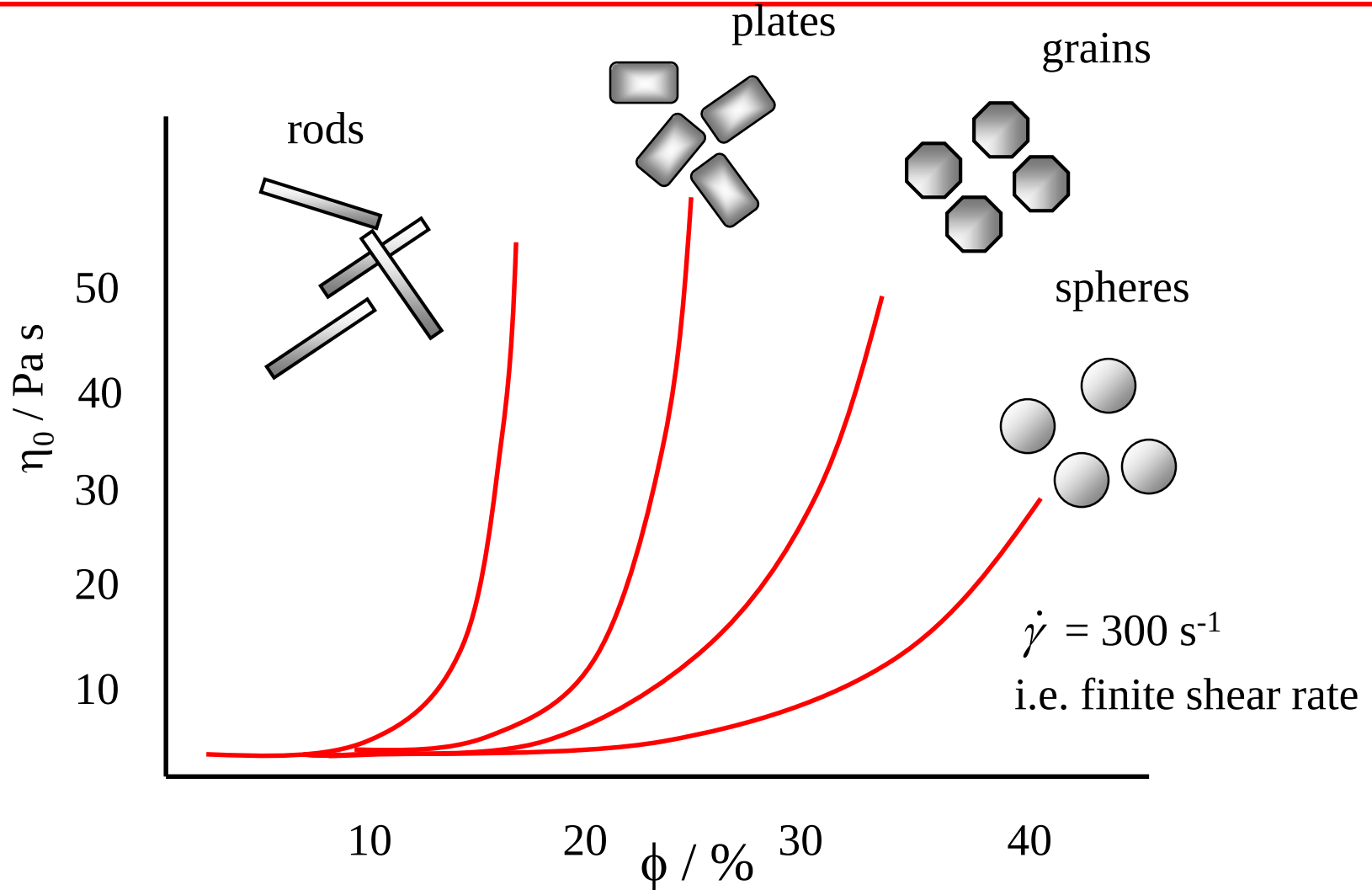


of

Particle Geometry		$[\eta]$	$\phi_{\max}$
Spheres: Simple cubic		2.50	0.5236
Body-centred cubic		2.50	0.6800
	Random close-packed	2.50	0.6392
Random loose-packed		2.50	0.5886
Rods or ellipsoid, random packed:			
Aspect ratio = 2		2.58	0.68
Aspect ratio = 10		5.93	0.42
Fibres , random packed:			
Aspect ratio = 7		3.80	0.374
Aspect ratio = 21		6.00	0.233

of

Influence of Particle Shape



$\phi_{\max}$ increases - rods > plates > grains > spheres

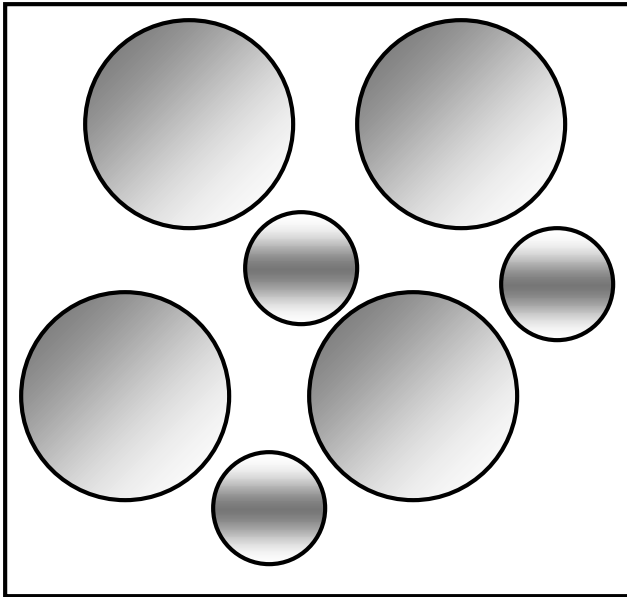
of

Effect of Polydispersity / Particle Size Distribution

Several studies carried out on bimodal systems

Polydispersity increases ϕ_{max}

(smaller particles fit in interstitial spaces between larger particles)



Often employed in emulsion polymer technology to increase solids at a given viscosity

$$\frac{\eta}{\eta_s} = \underbrace{\left(1 - \frac{\phi_1}{\phi_{1max}}\right)^{-[\eta_1]\phi_{1max}}}_{\text{fine fraction}} \underbrace{\left(1 - \frac{\phi_2}{\phi_{2max}}\right)^{-[\eta_2]\phi_{2max}}}_{\text{coarse fraction}}$$

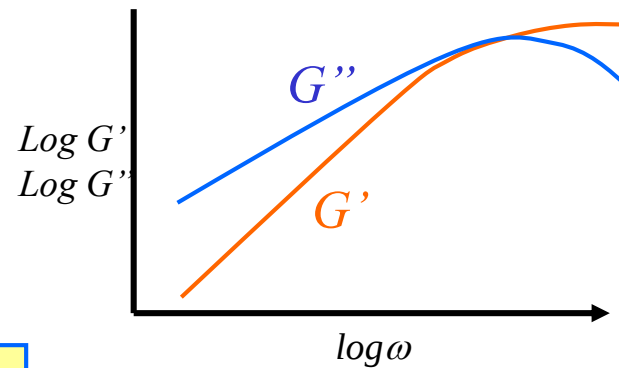
Systems with a complex, continuous distribution of particle sizes - impossible to predict ϕ_{max}

Maximum packing fraction must be determined experimentally

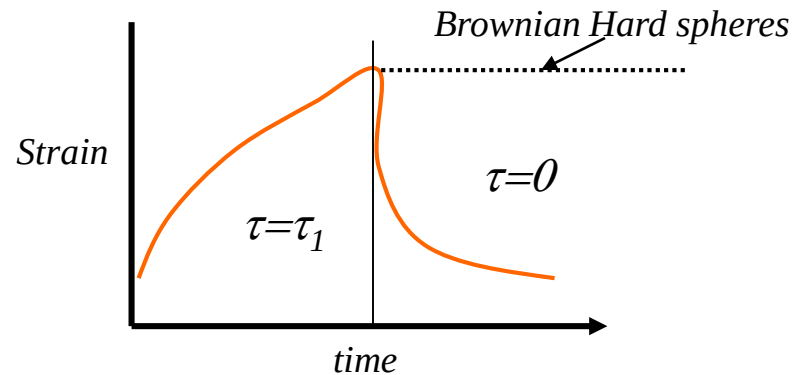
of

Brownian Hard Spheres cont'd

G, G'' of Brownian hard spheres:



Creep and creep recovery:



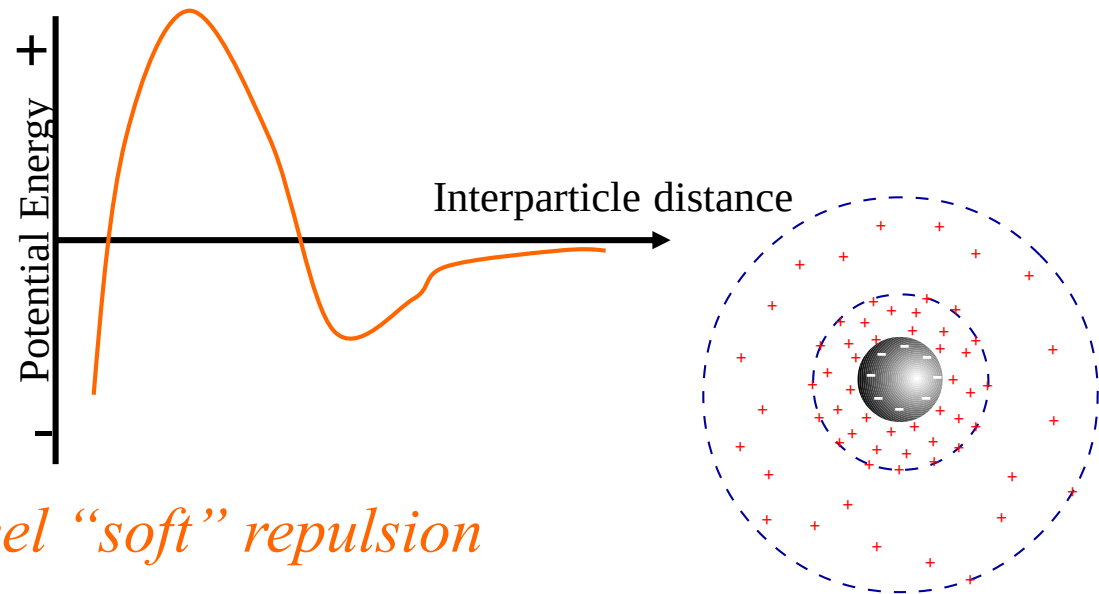
of

Stable Suspensions

Sterically or electrostatically stabilized

Electrostatically stabilized system

Submicron fillers where we have colloidal interactions



Particles will feel “soft” repulsion

of

Electrostatically Stabilized Suspensions

At low ϕ :

$$\eta = \eta_o \left(1 + \frac{5}{2} \phi + H_o \phi^2 \right)$$

H_o is a function of charges, etc.

Problem

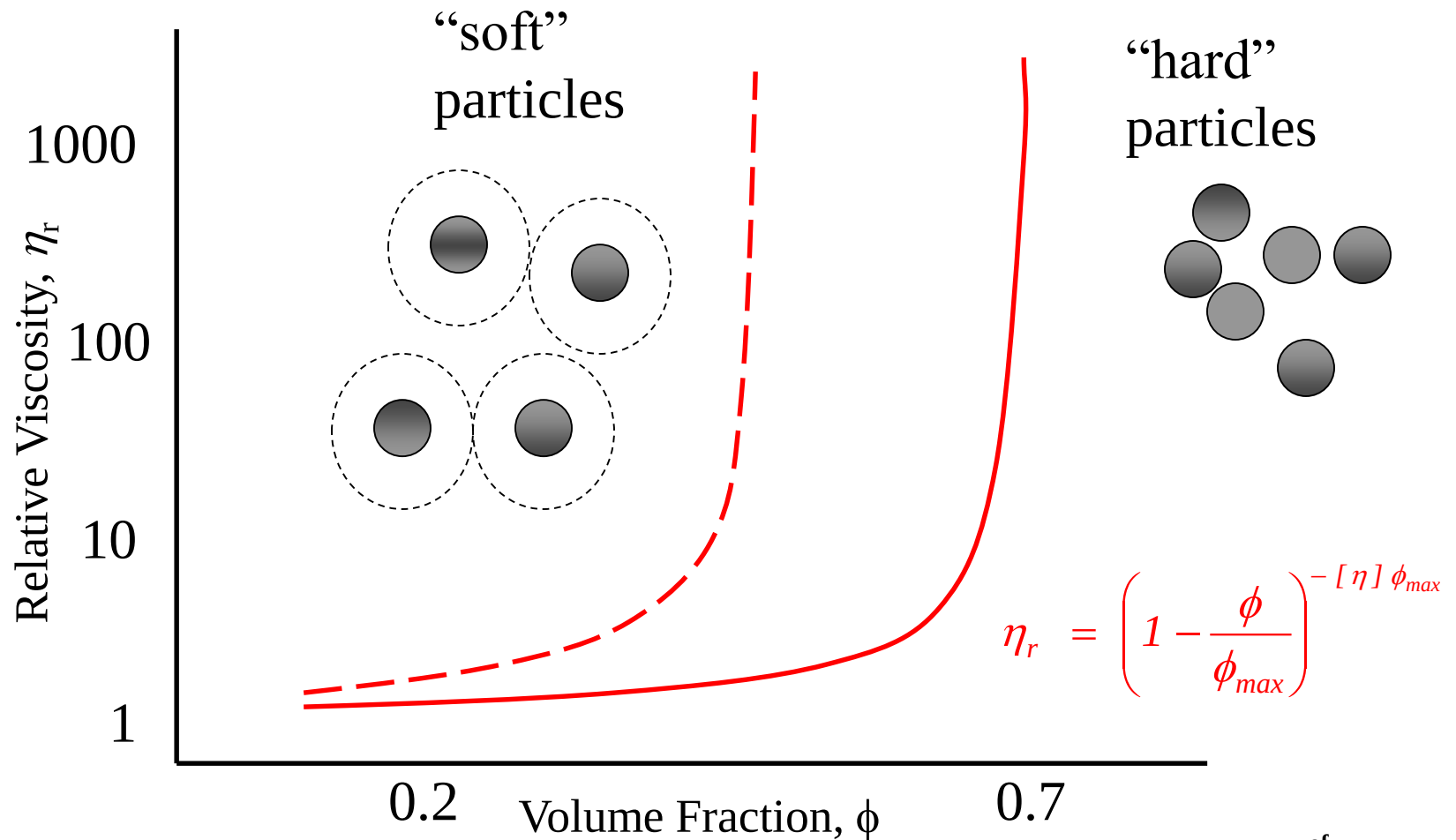
Have two systems at $\Phi=0.3$,

- electrostatically stabilized
- Brownian Hard sphere

Which will have a higher viscosity?

of

Viscosity vs Particle Volume Fraction – Influence of Colloidal Forces

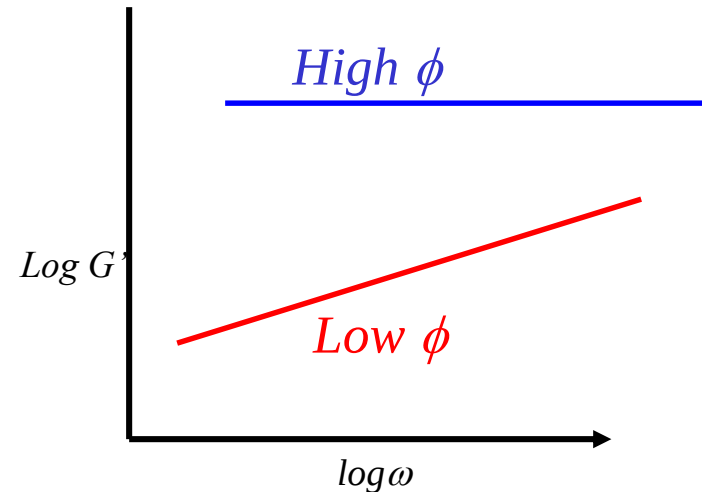


Electrostatically Stabilized Suspensions

Plot the dynamic frequency spectrum of G' for different ϕ

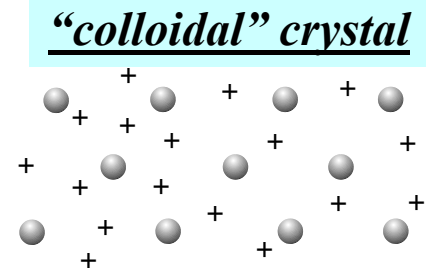
At low ϕ

G' frequency dependent and lower than G'' at low frequencies



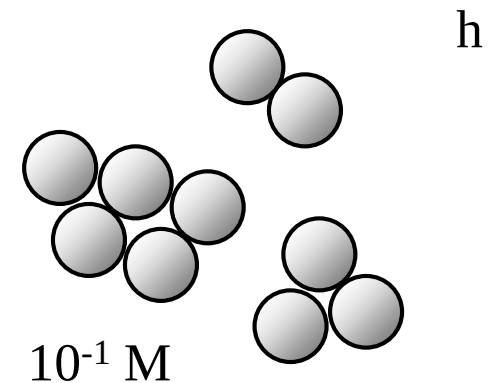
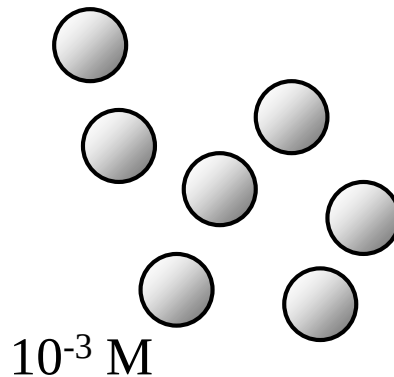
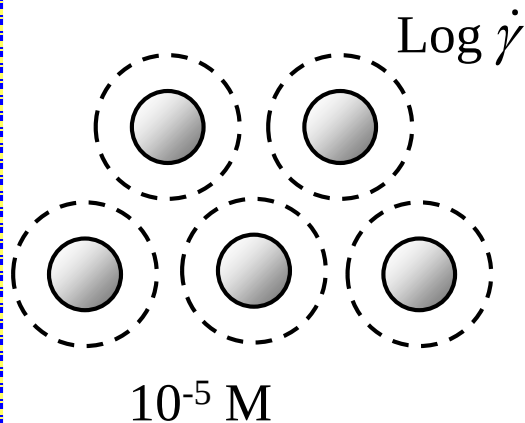
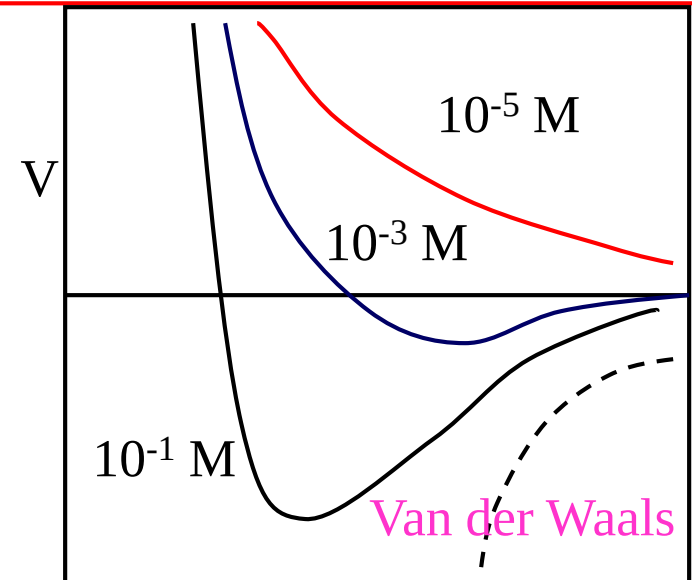
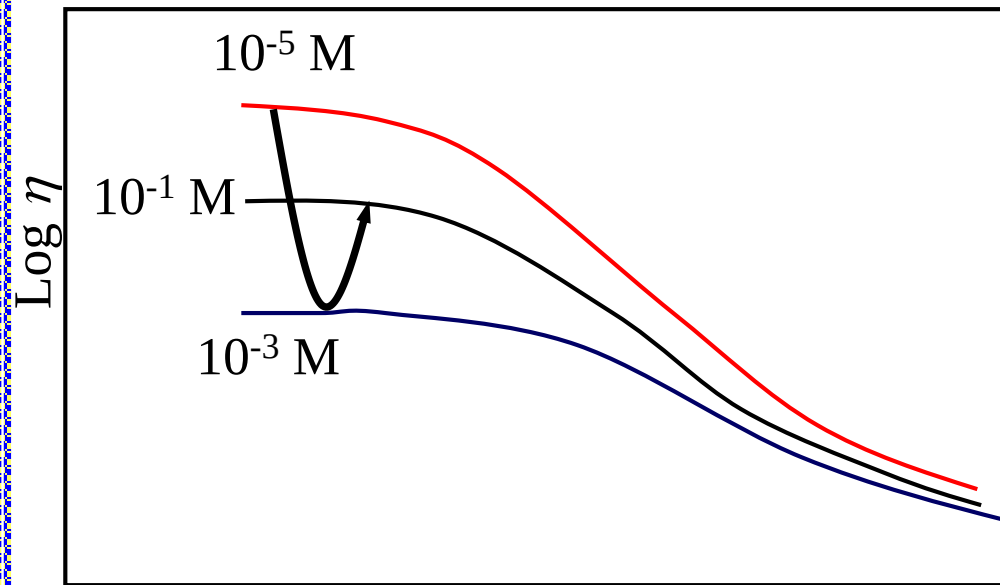
At high ϕ

Particles position themselves as far apart as possible in a lattice structure to minimize repulsion between them.



of

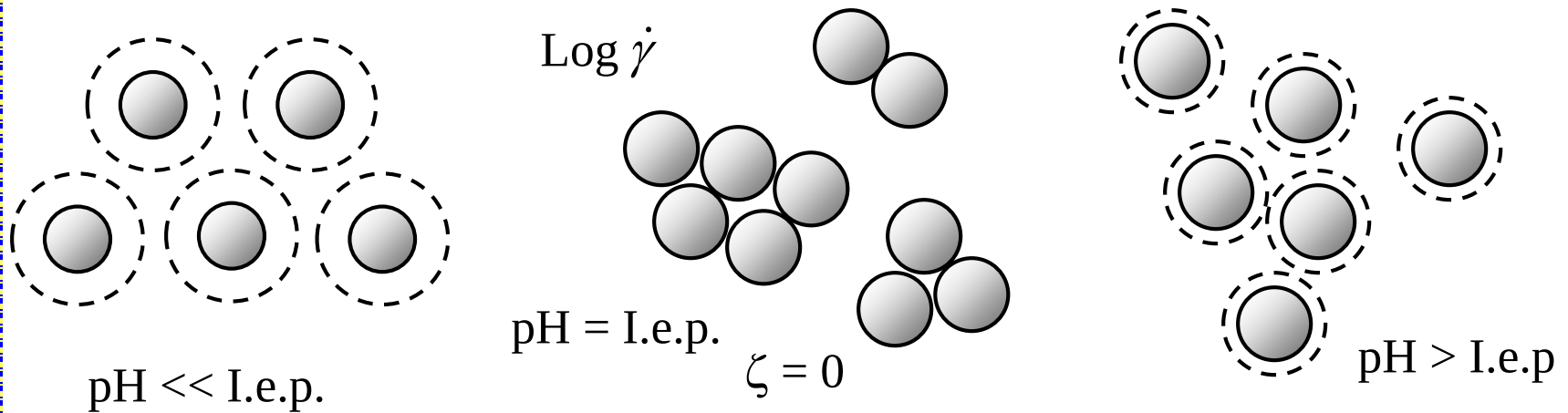
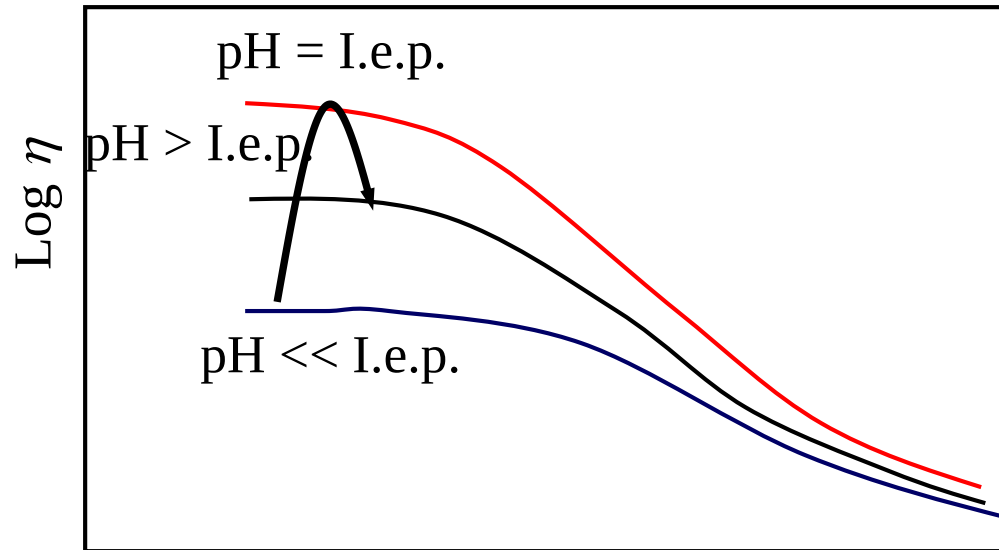
Particles with strong electrostatic repulsion: Electrolyte Effects



Increasing ionic strength →

of

Particles with strong electrostatic repulsion: pH Effects



Increasing pH →

of